

Youssef Naciri Zerta

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Currently at EMBL-EBI, Hinxton | Available for a full-time research placement, February–July 2027

RESEARCH STATEMENT

I work on representations of biology that recover mechanism rather than merely fit data, and on the evaluation machinery that decides which of the two you have - identifiability, leakage control, and calibration against real interventional measurements. In practice this means building pipelines around frozen frontier models of sequence and cellular state, designing the tasks that test what they actually encode, and holding the answer to a standard that survives adversarial review.

SELECTED RESEARCH

Cross-Modal Representation Diagnosis for Biological Foundation Models

2026

Independent research - in preparation, target: ICLR - *do protein and DNA models recover the same mechanism?*

- Testing whether protein and genomic foundation models recover the same latent representation of a mutation's functional effect, using endogenous saturation genome editing as interventional ground truth - a pre-registered test of the Platonic Representation Hypothesis in a setting where the central dogma supplies two views of one physical event.
- Built the frozen-encoder evaluation pipeline across **ESM2**, **ESM-C**, **Nucleotide Transformer v2**, and **HyenaDNA**: variant-resolved windowed embedding extraction under three poolings, cached row-aligned to a derived 3,893-variant table (reference-base check: 0 mismatches / 2,803 coding variants), with isolated environments reconciling mutually incompatible model stacks. Pipeline validated against a published zero-shot benchmark before any downstream analysis (Spearman $\rho = 0.496$ vs ≈ 0.50 reported).
- Result: the functional signal is *protein-private*, not shared - CCA subspace decomposition under position-grouped nested CV gives protein-private $\rho = 0.443$ vs shared 0.240 and DNA-private 0.102, while a DNA-only abundance signal loads on the DNA-private subspace exactly where biology requires. Protein models converge 3–5 $\times$ more than DNA models at matched family relatedness, calibrated against untrained-encoder floors and same-modality ceilings across four alignment metrics.
- Hardened and replicated: position-grouped CV retired a $\sim 3\times$ leakage-inflated DNA effect; the full result reproduces end-to-end on a second gene with byte-for-byte frozen analysis code and no retuning, strengthening rather than attenuating. Phase two in progress - naming what the shared and private subspaces encode, and cross-modal sparse-autoencoder feature correspondence. Frontier-scale DNA models (Evo 2) remain a stated scope limitation pending compute.

Causal Identifiability of Sparse-Autoencoder Features in Foundation Models

2026

Sole author - submitted, MLCB 2026 - *are interpretability features safe to read as mechanism?*

- A causally grounded test of whether interpretability “features” read out of a genomic foundation model are trustworthy, using saturation-mutagenesis MPRA for per-base-pair causal ground truth.
- Two-sided result: a real aggregate causal signal survives a full control suite, yet the per-feature SAE decomposition is non-identifiable - causal features barely reproduce across retrains, a second SAE family recovers a near-disjoint causal set from identical activations (cross-family Jaccard ≈ 0.01), and a single supervised direction stably decodes the effect. The failure is the decomposition, not the model.
- Generalizes across two SAE families (TopK, Gated) and two architectures (**HyenaDNA**, **Caduceus**). Implication: individual model features must not be read as mechanism without seed-stability and false-discovery control - directly relevant to model auditing and evaluation design.

RESEARCH EXPERIENCE

Lead Developer - AnnNet / SysBioVerse

2025–Present

EMBL-EBI, Saez Lab - *research infrastructure for biological network science*

- Architecting the core engine of **AnnNet**, a Python library for high-performance biological network science: heterogeneous topology (directed, signed, hyperedges, multilayer, condition-specific slices) on a single sparse incidence core, with typed annotation tables, lossless native storage, and lazy interoperability with NetworkX, igraph, graph-tool, and PyTorch Geometric.
- Designing the public API for topological representation and graph-based inference; adopted by groups at EMBL-EBI and Imperial College London, where standardized data structures keep signaling-network models reproducible across institutions.
- Growing AnnNet into SysBioVerse, an open, *scverse*-style ecosystem for systems and network biology.

Research Collaborator - Single-Cell Representation Learning

2025

- Co-developed an architecture combining contrastive learning and multi-head attention to learn patient-level embeddings from single-cell transcriptomics, reaching state-of-the-art results on disease-stratification benchmarks.
- Used cross-system transfer behavior as a direct probe of representation quality rather than reporting benchmark performance alone.

Undergraduate Researcher - Multiplex Bacterial Interaction Networks

2024–Present

Université Euromed de Fès - *first author, in preparation*

- Designed and ran a trait-resolved interaction study across 60 soil bacterial strains, generating 12 directed interaction layers from large-scale pairwise assays - data produced at the bench, not downloaded.
- Built a multilayer network pipeline (community detection, motif analysis, node-role characterization) over the resulting 3D tensor, yielding a 4-dimensional ecological embedding that classifies microbial strategies across antagonistic, metabolic, and morphogenetic modes.

PUBLICATIONS & PREPRINTS

- **Y. Naciri Zerta.** *A Causal Signal Without Causal Features: identifiability of sparse-autoencoder features in genomic foundation models.* Sole author, submitted (MLCB 2026). [\[preprint\]](#)
- *Great patients embed alike: contrastive learning for sample representation from single-cell data.* [\[preprint\]](#)
- *Cross-modal representation diagnosis for biological foundation models.* In preparation (ICLR).
- *AnnNet: a unified annotated-graph structure for biological networks.* In preparation (Bioinformatics).
- *Patterns of pairwise interactions between Streptomyces spp* In preparation (ISMEJ).

EDUCATION

State Engineer (Diplôme d'Ingénieur d'État) - Biotechnology

Expected June 2027

Université Euromed de Fès (UEMF), Morocco - *five-year engineering degree; Master's-equivalent, 300 ECTS*

Coursework: Metabolic Engineering, Recombinant DNA Technology, Advanced Microbiology, Viral Genetics, Optimization, Probability & Statistics.

TECHNICAL SKILLS

Foundation Models: ESM2, ESM-C, Nucleotide Transformer, HyenaDNA, Caduceus, Evo - embedding extraction, frozen-encoder pipelines, probing, domain adaptation, evaluation-task design.

Representation Learning: contrastive learning, graph neural networks, subspace and canonical-correlation methods, sparse autoencoders, transfer evaluation.

Evaluation & Statistics: nested and group-structured cross-validation, leakage diagnosis, false-discovery control, floor/ceiling calibration, pre-registration, replication design.

Computational Biology: variant-effect prediction, saturation genome editing / DMS / MPRA, single-cell genomics, multi-omics integration, network and mechanistic modeling.

Engineering: Python (PyTorch, NumPy, scikit-learn), C/C++, R, Bash, Docker, Google Cloud Platform; CUDA environment isolation, cached array pipelines, library and API design.

Wet Lab: recombinant DNA technology, genome engineering, molecular cloning, high-throughput bacterial culture.

Languages: English & French (fluent), Arabic & Tamazight (native), Spanish (basic).